

Rotation-Invariant Positive Kernels Defeat Two Abstract Rigidity Questions

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Status. Candidate full counterexamples, likely valid, to Questions 5.1 and 5.4 in the source paper.

1 Source questions

Bilokopytov calls a nondegenerate kernel K projectively invariant under a group action if, for every group element g , the transformed kernel $K(g \cdot z, g \cdot w)$ is a rescaling of K . A multiplier is a nonvanishing function implementing that rescaling; it is naturally determined only up to a constant unimodular factor.

Question 5.1 asks whether two nondegenerate positive semidefinite continuous kernels on a transitive topological G -space, both projectively invariant and having equal multipliers, must satisfy $L = cK$ for some $c > 0$ [1].

Question 5.1. *Let X be a set (topological space) and let K and L be non-degenerate positive semi-definite (continuous) kernels on X . Assume that a (topological) group acts on X transitively (and continuously) and both K and L are projectively invariant with respect to this action. Is it true that if they have equal multipliers, then there is $c > 0$ such that $L = cK$?*

Question 5.4 asks whether a separately continuous, positive semidefinite, nondegenerate kernel must be constant if G acts transitively and continuously and

$$K(g \cdot x, g \cdot y) = e^{\alpha(g)} K(x, y)$$

for a continuous homomorphism $\alpha : G \rightarrow \mathbb{R}$ [1].

Finally let us ask if an abstract version of Lemma 3.12 holds.

Question 5.4. *Let X be a topological space and let K be a separately continuous positive semi-definite non-degenerate kernel on X . Assume that G is a group that acts transitively and continuously on X and there is continuous homomorphism $\alpha : G \rightarrow \mathbb{R}$, such that $K(g \cdot x, g \cdot y) = e^{\alpha(g)} K(x, y)$, for any $x, y \in X$ and $g \in G$. Does it follow that K is a constant?*

2 Idea of the proof

Positive definiteness is compatible with a large supply of invariant kernels: every unitary representation gives matrix-coefficient kernels. The smallest connected example already suffices. On

the circle, the two features 1 and $\sqrt{a}z$ give the positive kernel $K_a(z, w) = 1 + az\bar{w}$. Rotating both variables cancels the two phase factors, so every member of the family has the same trivial multiplier. Changing a changes the kernel without changing the symmetry data. This is exactly the freedom that sesqui-holomorphic rigidity removed in the theorem preceding Question 5.1.

3 Counterexample theorem

Theorem 1. *Let*

$$X = G = \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}, \quad g \cdot z = gz,$$

with their usual topologies. For $0 < a < 1$, define

$$K_a(z, w) = 1 + az\bar{w}.$$

Then:

1. K_a is a continuous, nondegenerate, positive semidefinite kernel, is nowhere zero, and is exactly G -invariant;
2. for every $g \in G$, the complete family of multipliers rescaling $K_a(g \cdot z, g \cdot w)$ to $K_a(z, w)$ consists of the constant unimodular functions, independently of a ;
3. if $0 < a < b < 1$, then K_b is not a positive scalar multiple of K_a .

Consequently, (K_a, K_b) is a counterexample to Question 5.1. Moreover, (X, G, K_a, α) with $\alpha \equiv 0$ is a counterexample to Question 5.4.

Proof. The rotation action is continuous and transitive. For arbitrary $z_1, \dots, z_N \in \mathbb{T}$ and $c_1, \dots, c_N \in \mathbb{C}$,

$$\sum_{i,j=1}^N c_i \bar{c}_j K_a(z_i, z_j) = \left| \sum_{i=1}^N c_i \right|^2 + a \left| \sum_{i=1}^N c_i z_i \right|^2 \geq 0.$$

Thus K_a is positive semidefinite. It is continuous, and $K_a(z, z) = 1 + a > 0$, so it is nondegenerate in the source's sense. Also

$$|K_a(z, w)| = |1 + az\bar{w}| \geq 1 - a > 0,$$

so it never vanishes.

For $g \in \mathbb{T}$,

$$K_a(gz, gw) = 1 + a(gz)\overline{gw} = 1 + az\bar{w} = K_a(z, w).$$

Hence K_a is projectively invariant, and the constant function 1 is a multiplier for every rotation.

We also verify the stronger, convention-independent statement about the whole multiplier class. Suppose a nonvanishing function ω rescales the transformed kernel to the original one. Exact invariance gives

$$\omega(z)\overline{\omega(w)}K_a(z, w) = K_a(z, w) \quad (z, w \in \mathbb{T}).$$

Since K_a is nowhere zero, $\omega(z)\overline{\omega(w)} = 1$ for all z, w . Taking $z = w$ gives $|\omega(z)| = 1$, and then varying z, w shows that ω is constant. Conversely every constant unimodular function works. The multiplier class therefore does not depend on a .

If $K_b = cK_a$ for $c > 0$, evaluating at $(z, w) = (1, 1)$ and $(1, -1)$ yields

$$1 + b = c(1 + a), \quad 1 - b = c(1 - a).$$

Eliminating c gives $(1+b)(1-a) = (1-b)(1+a)$, hence $2(b-a) = 0$, contrary to $a < b$. This proves the counterexample to Question 5.1.

Finally take any $0 < a < 1$ and $\alpha(g) = 0$. This is a continuous homomorphism $\mathbb{T} \rightarrow \mathbb{R}$, and exact invariance gives the covariance identity in Question 5.4. But

$$K_a(1, 1) = 1 + a \neq 1 - a = K_a(1, -1),$$

so K_a is not a constant function on $X \times X$. Question 5.4 is therefore also answered negatively. $\square$

4 Verification and robustness

The proof checks every printed hypothesis directly. The same example works on a connected compact space with a connected compact acting group, so the failure is not caused by discreteness. Restricting to $0 < a < 1$ is not needed for positive semidefiniteness, but it makes the kernel nowhere zero and removes any ambiguity about the full multiplier class.

The script `code/verify_rotation_kernels.py` samples finite Gram matrices, rotations, and scalar-multiple tests for several parameters. It is only a sanity check; positive semidefiniteness and all identities are proved symbolically above. The independent audit is recorded in `VERIFICATION.md`.

5 Scope and novelty check

This packet settles only the two abstract Questions 5.1 and 5.4. It does not answer the paper’s Questions 5.2 or 5.3, nor its broader classification of automorphism-invariant pull-back metrics on complex domains.

A bounded search on 2026-08-13 used arXiv:1711.05211, the exact title and author, the exact phrases “equal multipliers” and “does it follow that K is a constant”, and projectively invariant positive-kernel keywords. It found the source, the author’s dissertation repeating the surrounding program, and general literature on projectively invariant kernels, but no later source explicitly answering Questions 5.1 or 5.4. Novelty confidence is moderate; mathematical-validity confidence is high because the example is elementary and all assumptions are verified directly.

References

- [1] E. Bilokopytov, *Invariant Hermitean Metric Generated By Reproducing Kernel Hilbert Spaces*, arXiv:1711.05211v3 (2018), Questions 5.1 and 5.4.